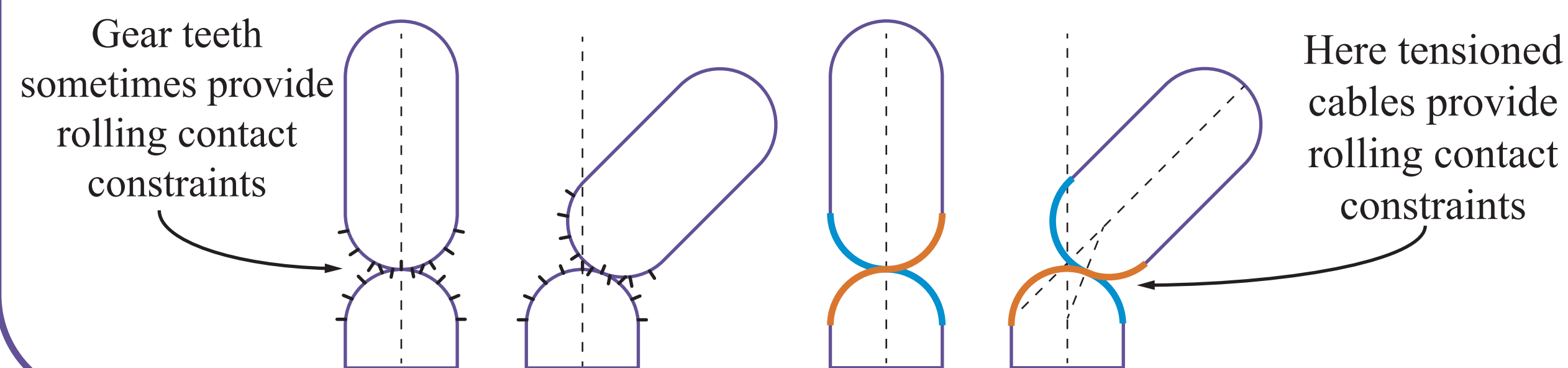


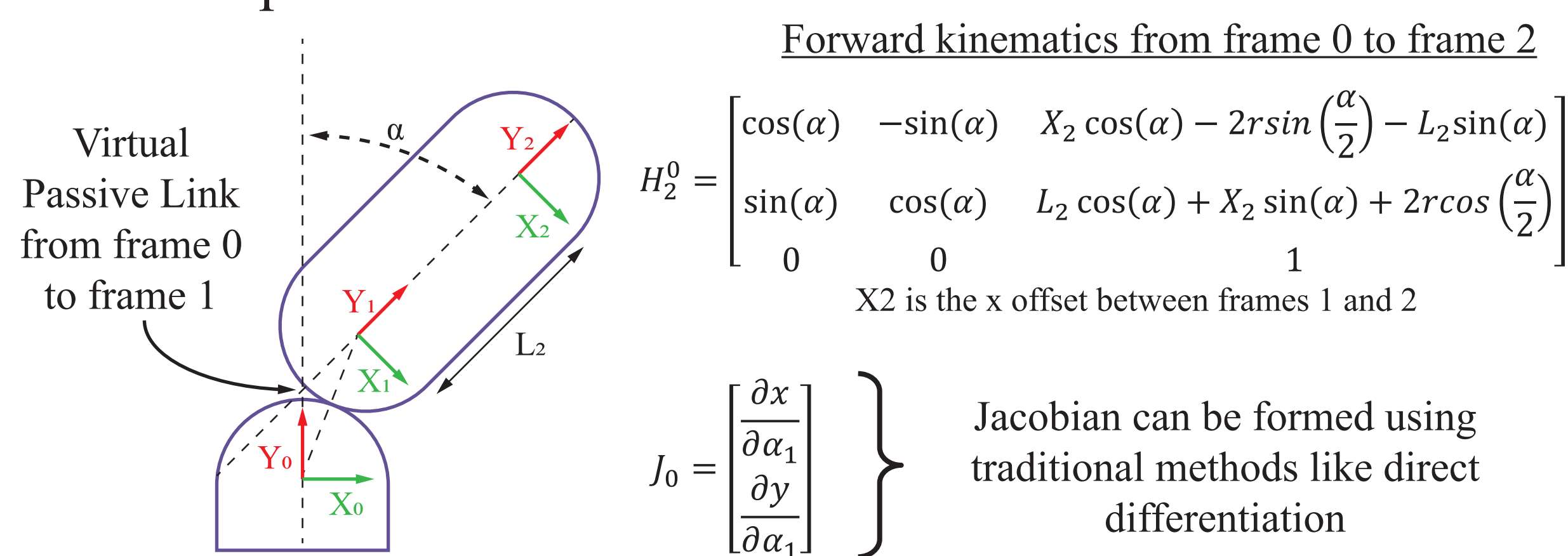
Motivation:

Remotely locating actuators away from a manipulator's driving point and joints is advantageous in terms of the robot's form factor and inertia **but complicates control**.

Rolling contacts use cables, interlocking teeth, or contact friction to make a low friction joint.



Forward kinematics of rolling contacts effectively adds a "virtual passive link".



N+1 control relies on tendon tension control at the actuator level. It relates tendon tensions to joint torques via a tendon routing matrix or jacobian.

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{21} & R_{22} & R_{23} & R_{24} \\ R_{31} & R_{32} & R_{33} & R_{34} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}$$

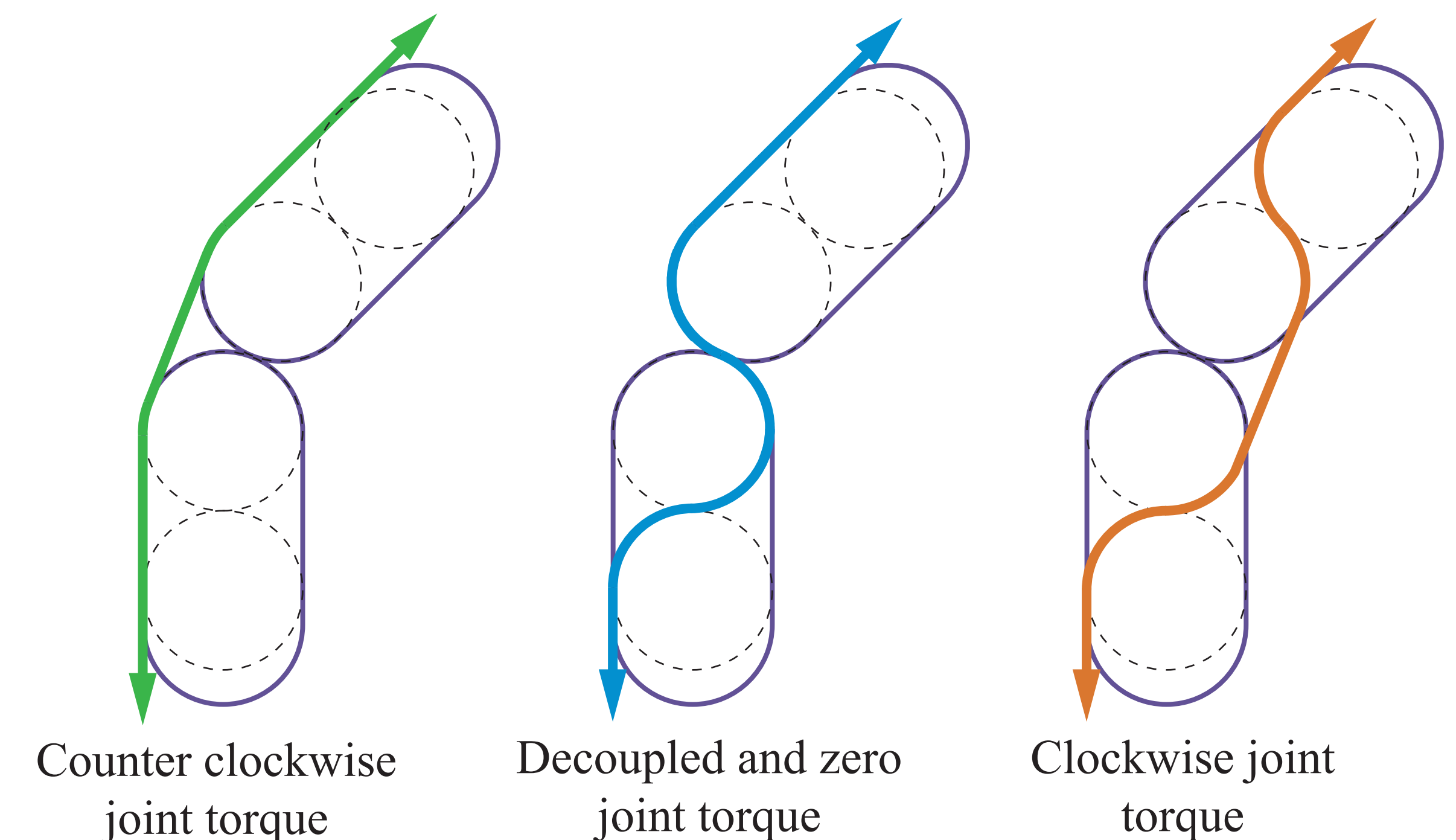
Calculating joint torques from tendon tensions is generally easy.

Solving tensions from torques is hard because of an inherent null space and constrained nature (positive tensions only)

Solution:

N+1+Rolling Contacts enables full control of a robot's joints while simplifying control.

Tendons routed through the center of the joints result in zero moments about the joint, selectively eliminating joint coupling.



Rolling contacts allow calculating minimum tendon tensions from desired joint torques in closed form.

Tendon tensions to Joint Torques

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \begin{bmatrix} R_1 & -R_1 & 0 & 0 \\ R_2 & 0 & -R_2 & 0 \\ R_3 & 0 & 0 & -R_3 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}$$

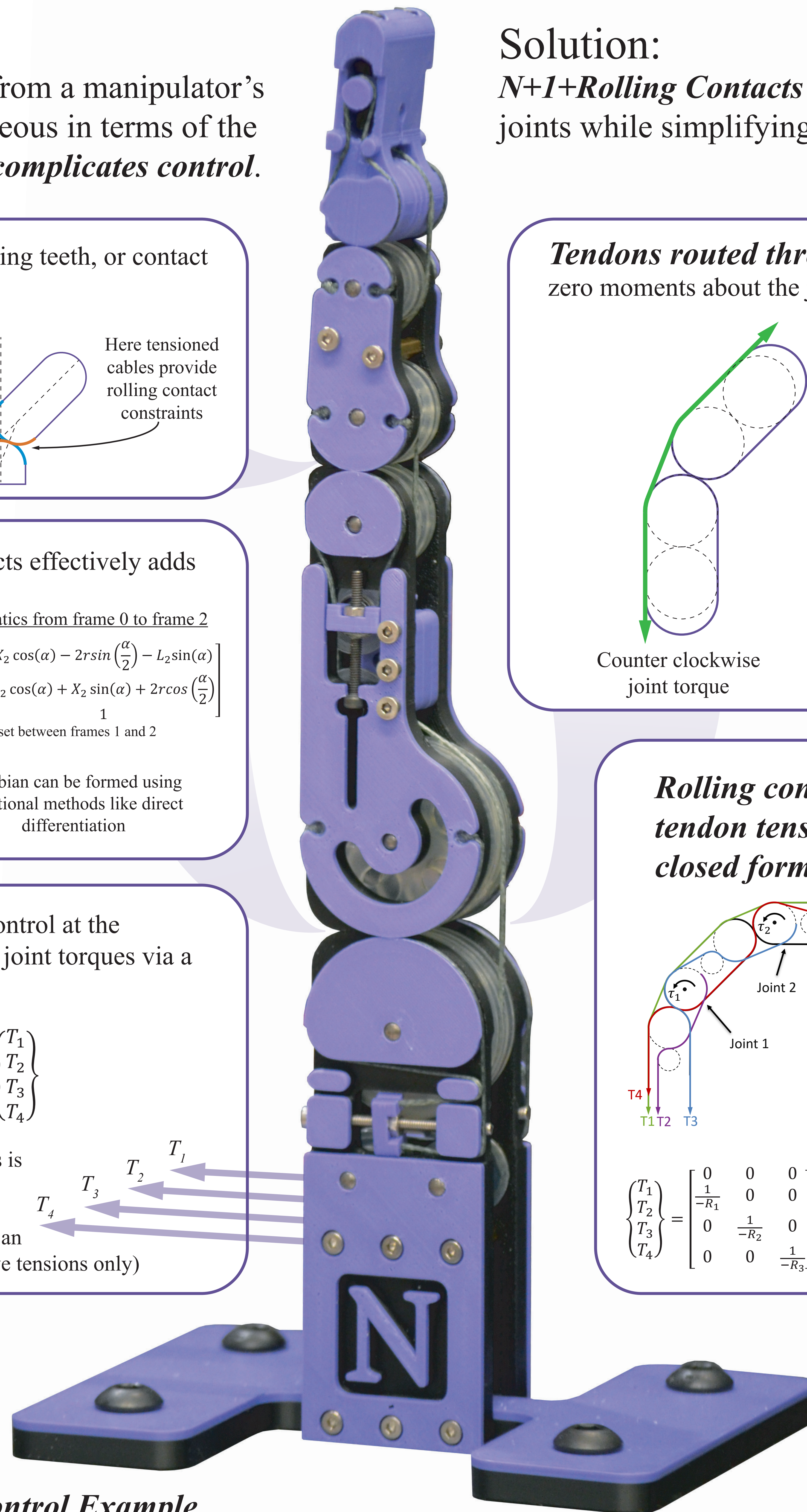
Desired Joint Torques to Tendon Tensions

$$T_1 = \max(0, \frac{\tau_1}{R_1}, \frac{\tau_2}{R_2}, \frac{\tau_3}{R_3})$$

$$T_2 = \frac{R_1 T_1 - \tau_1}{R_1} \quad T_3 = \frac{R_2 T_1 - \tau_2}{R_2} \quad T_4 = \frac{R_3 T_1 - \tau_3}{R_3}$$

$$\begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{-R_1} & 0 & 0 \\ 0 & \frac{1}{-R_2} & 0 \\ 0 & 0 & \frac{1}{-R_3} \end{bmatrix} \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \max(0, \frac{\tau_1}{R_1}, \frac{\tau_2}{R_2}, \frac{\tau_3}{R_3})$$

Set minimum tendon tension preload greater or equal to zero



Operational Space Impedance Control Example

High sample rates are needed to render stiff virtual environments or to produce high gain controllers. Running an iterative optimization to solve a typical inverse tendon routing problem limits the maximum sample rate of an impedance control loop. Comparable closed form solutions don't minimize tendon tension and instead use a heuristic.

